

NEW PLYMOUTH, New Zealand

WMO: 933090

Lat: 39.0078S

Lon: 174.1836E

Elev: 28

StdP: 100.99

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.3	3.5	-0.1	3.7	3.8	0.9	4.1	5.0	15.4	12.0	14.3	11.8	3.1	140	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.3	23.8	19.1	22.8	18.8	21.9	18.4	20.6	22.3	19.9	21.6	19.2	21.0	5.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	14.7	21.5	19.2	14.0	20.9	18.5	13.4	20.4	59.5	22.3	57.0	21.7	54.9	21.1	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.4	11.9	10.7	-0.2	26.2	1.0	1.5	-0.9	27.3	-1.5	28.2	-2.0	29.0	-2.7	30.2
			-0.8	21.5	0.9	1.0	-1.5	22.3	-2.0	22.8	-2.6	23.4	-3.2	24.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	17.5	18.1	16.8	14.7	12.8	10.7	9.8	10.2	11.5	12.5	13.9	16.1	
	DBStd	3.50	2.15	1.97	2.12	2.23	2.42	2.50	2.22	2.00	1.97	1.91	1.96	2.08	
	HDD10.0	93	0	0	0	1	5	22	32	22	9	3	0	0	
	HDD18.3	1742	41	26	57	109	172	229	264	251	206	180	134	73	
	CDD10.0	1441	233	226	209	143	92	43	26	29	53	81	116	189	
	CDD18.3	49	16	19	8	1	0	0	0	0	0	0	0	4	
	CDH23.3	52	20	23	5	0	0	0	0	0	0	0	1	4	
CDH26.7	1	1	0	0	0	0	0	0	0	0	0	0	0		
Wind	WSAvg	5.2	5.0	4.7	4.9	4.7	5.1	5.3	5.2	5.1	5.5	5.9	5.7	5.2	
Precipitation	PrecAvg	1439	101	93	102	117	137	149	152	133	123	122	115	107	
	PrecMax	2016	234	398	362	337	262	270	298	277	249	448	241	190	
	PrecMin	1120	16	4	13	23	11	52	11	53	31	19	44	20	
	PrecStd	183	56	68	65	69	49	48	66	50	55	73	51	45	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	25.5	24.0	21.4	19.2	16.6	15.6	15.6	17.1	18.5	21.4	23.2	
		MCWB	20.1	20.0	18.7	17.6	16.5	15.1	13.8	13.2	14.0	15.3	16.7	18.6	
	2%	DB	23.6	24.0	22.6	20.1	18.2	15.8	14.7	14.8	15.9	17.1	19.5	21.7	
		MCWB	19.3	19.3	18.3	17.1	16.2	14.4	13.1	12.8	13.7	14.7	16.1	17.8	
	5%	DB	22.6	22.9	21.7	19.4	17.4	15.3	14.2	14.2	15.2	16.4	18.4	20.7	
		MCWB	18.9	19.1	18.0	16.7	15.7	13.8	12.5	12.4	13.3	14.2	15.6	17.4	
	10%	DB	21.6	22.0	20.8	18.7	16.8	14.7	13.6	13.7	14.7	15.7	17.4	19.8	
		MCWB	18.4	18.6	17.6	16.2	15.1	13.1	11.9	11.9	13.0	13.8	15.0	17.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.5	21.7	20.5	19.2	17.6	15.9	14.7	14.4	15.2	16.1	18.0	20.1
			MCDB	23.6	23.3	22.0	20.2	18.2	16.3	15.0	14.9	15.8	17.2	19.7	21.5
2%		WB	20.4	20.9	19.7	18.3	16.9	14.9	13.7	13.6	14.6	15.4	17.0	19.1	
		MCDB	22.3	22.4	21.2	19.3	17.6	15.5	14.5	14.2	15.3	16.5	18.6	20.5	
5%		WB	19.6	20.2	19.0	17.6	16.2	14.2	12.9	13.0	14.1	14.9	16.3	18.4	
		MCDB	21.5	22.0	20.8	18.8	17.2	15.1	13.8	13.8	14.8	15.9	17.7	19.9	
10%		WB	19.0	19.4	18.2	16.7	15.3	13.4	12.1	12.3	13.5	14.3	15.6	17.7	
		MCDB	20.9	21.4	20.2	18.2	16.6	14.5	13.3	13.4	14.5	15.4	17.0	19.3	
Mean Daily Temperature Range		5% DB	MDBR	6.9	7.3	7.5	7.1	6.3	6.0	6.2	6.2	5.9	5.9	6.3	6.2
			MCDBR	7.9	7.9	8.1	7.3	5.8	4.8	5.7	6.1	6.1	6.2	7.3	7.3
	MCWBR		5.0	4.9	5.2	5.1	4.6	4.2	4.8	4.7	4.8	4.9	5.1	4.8	
	5% WB	MCDBR	6.3	6.1	6.1	5.6	4.3	4.1	4.6	4.7	4.7	5.0	5.6	5.5	
		MCWBR	4.5	4.6	4.6	4.7	4.0	4.1	4.4	4.3	4.3	4.4	4.6	4.2	
Clear-Sky Solar Irradiance	taub	0.347	0.342	0.325	0.313	0.305	0.300	0.295	0.309	0.335	0.338	0.335	0.351		
	taud	2.540	2.575	2.621	2.646	2.636	2.626	2.638	2.599	2.507	2.512	2.538	2.529		
	Ebn at Noon	984	965	942	892	837	808	840	882	914	958	990	985		
	Edh at Noon	108	100	88	75	65	61	64	76	96	105	108	110		
All-Sky Solar Radiation	RadAvg	6.88	6.02	4.83	3.26	2.18	1.70	1.97	2.76	3.85	5.08	6.26	6.52		
	RadStd	0.41	0.54	0.24	0.23	0.12	0.12	0.15	0.17	0.28	0.36	0.49	0.44		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	+0.43	N/A	N/A	N/A	-89	N/A	N/A
(47)	Regional (4 neighbors)	+0.29	+0.50	+0.61	N/A	N/A	N/A	-31	-104	+79

Nomenclature: See separate page